

University of Essex

MSc Artificial Intelligence

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Unit 5: Collaborative Discussion – Agent Communication Languages

Initial Discussion

The discussion at hand revolves around two topics, each addressed in turn:

1. What are the potential advantages and disadvantages of the use of agent communication languages such as KQML?
2. How do they compare with method invocation in Python or Java?

For the first topic, I would assert that KQML offers the advantages of an asynchronous, content-agnostic, declarative approach. Autonomous agents trade messages with each other based on intent, rather than providing specific details about implementation of that intent (Das et al, 2023). In other words, agents don't need to know about the internal workings of other agents, and communicate with each other using directives for action (e.g., "do this task") that are independent of how the task will be accomplished in terms of runtime code or data manipulation (Das et al, 2023). A significant drawback, however, is that debugging distributed agent communications can quite complex, especially given the limited tools currently available (Das et al, 2023).

For the second topic, my view is that method invocations within code take an orthogonal, if not diametrically opposite, approach, as they are usually synchronous and tightly coupled to the data they work with. Strongly typed languages such as Java, in particular, are far less flexible than KQML in that they depend on all actors within the operational runtime using the same data typing schemas in a manner compatible with Java's data processing capabilities and limitations (Lelek and Skeet, 2022).

References

Das, A., Chen, S.-C., Shyu, M.-L. and Sadiq, S. (2023). Enabling Synergistic Knowledge Sharing and Reasoning in Large Language Models with Collaborative Multi-Agents. [online] doi:<https://doi.org/10.1109/cic58953.2023.00021>.

Lelek, T. and Skeet, J. (2022). *Software Mistakes and Tradeoffs: How to Make Good Programming Decisions*. Simon and Schuster.